

Safe System of Work

Use of Ride on Roller

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Overview

The following information provides guidance on the company's Standard Operation Procedures for works involving the use of a Ride on Roller

All employees have a responsibility for their safety and the safety of others by adhering to these codes of practice.

Ride on rollers purpose is to uniformly compact the ground and leave a smooth, level or cambered surface. Different types of roller, vibratory rollers, dead weight rollers, wheeled or metal drummed rollers etc.

There are two aims of compaction and they are to remove the air voids (empty spaces) and minimise settlement. The word consolidation means the removal of water to increase soil density.

Dead weight rolling - this is the weight of the roller alone providing the compaction of the material.
Vibratory Roller- this is where the roller uses a vibratory system as well as the weight of the roller to compact the material. A vibratory pass is 3 x more effective than a dead weight pass.

Hazards Identified

The use of a Ride on Roller could result in collisions with people or vehicles or machine rollovers, these can result in serious injury or can possibly be fatal.

- Adverse weather
- Cuts and abrasions
- Vehicle movements
- Collisions
- Falls from height
- Slips & Trips

Personal Protective Equipment (PPE) Requirements



Pre-Use Checks

First walk round the roller checking for any damage to the machine making sure there are no leaks or damage, making sure the scrapper bars are in the correct position from the rollers. Check that all the lights are in good condition and clean. Check that the ROPS (Roll Over Protective Structure) is in good condition with the pins and clips secured and in place.

Engine compartment check:

- Engine oil level
- Hydraulic oil level
- Coolant level
- Fuel level
- Brake fluid level
- Air filter indicator

Running Checks

Get onto the roller and put on the safety belt and adjust seat as needed. Make sure the lever is in neutral with the brake on. Start the engine and let it warm up for about 2 to 5 minutes. Check that all the lights and the horn is working, check that the vibratory system, steering, braking and movement of the roller is in good working order. Fill out the daily check sheet.

Safety:

When checking the oil level always wear gloves, this can prevent skin diseases and prevent contamination of oil onto operating controls and cab.

If topping up the hydraulic/ fuel system make sure you have a clean can with a funnel with a filter in, clean around the filler cap so no dirt can get into the system.

If the engine is hot do not remove the radiator or expansion tank cap because the system is pressurised and could cause hot water/ steam to come out and burn the operator.

General Safety

Ensure you are trained to operate the plant & equipment, Qualifications & Certificates offer credibility, proof of skills, employment prospects and promotional prospects.

Always read the operators manual, the manual should be on the machine or easy access to it if you

need to find something out. Other personnel that may need to use the manual are supervisors, planners, and maintenance staff, low loader drivers, if removed replace as soon as possible.

Always face the machine using three points of contact (two feet and one hand at all times) when climbing on or off. **Never jump.**

A hard hat on construction sites must always be worn unless you are sitting in an enclosed cab that meets the falling objects protective structure (FOPS) criteria.

Wear the seat belt if fitted as in the event of a roll over keeps the operator within the confines of the operating seat which **may** minimise injury by not being flung around.

Do not carry passengers on the machine unless a seat is provided. Be aware of people entering the working area of the machine.

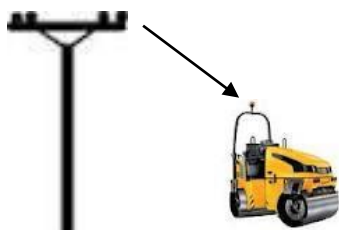
Never leave machine unattended with the engine running.

Do not strike/run over kerbs.

Do not change direction without stopping the vibration and allowing the machine to come to rest.

Do not stop any movement with a jerk as this will result in a ripple in the surface.

When travelling near open trenches make sure the roller is more than the depth of the trench away (2m trench, stay 2m away) because if not it could cause the sides of the trench to collapse.



The distance you must be away from the powerlines will be determined by the voltage of the electrical lines. This can be determined by its support, if it's on wooden posts or metal pylons. This is important because certain weather conditions electric can arc (jump).

Operations

Before starting work on the compacting area check that there are no obstacles in the way, the area is safe, no pedestrians, no underground services, method statement and risk assessment have been read and understood, know the start and finishing points of the runs and the compaction specifications.

A rolling pass is generally accepted as compacting in one direction and returning back on the same path to the start point (up and back). An operator needs to know how many rolling passes to make so the material is compacted to the desired level.

Factors to be considered that determine the number of rolling passes:

- deadweight or vibration rolling
- type of material
- weight of the machine
- moisture content
- thickness of the material

The most common cause for poor compaction is, to compact a too THICK a layer of material. Soils should be compacted in layers of approximately 200mm.

If rolling a cross fall or camber the first pass should be made on the low side and work up so that the loose material does not roll downwards.

Always manoeuvre the roller to make the next pass on already compacting ground with an overlap of from 150mm to half the width of the roller depending on the material.

Do not stop/change direction at the same place on every pass. Stager the stopping place to prevent transverse roller marks

Rollers are heavy smooth drum machines that are designed to compact the material so if the roller was to be driven off the work area it could damage the ground it moves on, get stuck, damage the machine and even damage underground services.

Rollers can have both drums driven or one drum pushing/ pulling the other drum Un-driven rollers form a slight 'bow wave' in front of the roll.

Driven rollers leave a slight 'bow wave' behind the roll when movement is initiated, and/or when direction is changed.

Dynamic or vibration rollers must have the vibration stopped when stationary as they would tend to sink into the surface leaving an indentation and possibly 'bow waves' on both sides on the roller.

Dynamic or vibration rollers must have the vibration stopped when on hard surfaces or damage may be sustained by the machine.

Sharp turns will result in scuffing the surface.

When rolling Tarmac or Bitumen material with a dead weight roller the drive roller should lead on to uncompacted material so as not to create a bow wave

Slopes

If the roller has to be travelled up a slope the operator must:

- make sure the gradient of the slope is within the rollers limits
- sufficient grip on the surface for the rollers to travel on
- no obstacles
- safe area/ firm ground

If compacting on a gradient roll up and down the slope and not across because the machine can become unstable and tip over.

Material

The material generally used as a sub- base is a gravel and sand mix (803)

If the operator has to roll a granular based material in thin layers, then a roller with low frequency amplitude would generally be used.

When rolling Tarmac or Bitumen material the water sprinkler system must be on to prevent the material from sticking to the drums.

Stability

Possible causes of making the roller tip over

- Driving too fast for the ground conditions
- Turning to sharply
- To steep slopes or driving across slopes
- Uneven ground/ trenches, potholes in the ground
- Soft ground

Soft, wet clay/ground

This could cause the roller to lose grip, slide, get stuck and even turn over

Most Ride on Rollers have an articulated chassis that swivels or pivots in the centre. One pivot plane or angle (looking from the top) provides steering (articulation) and the other one looking from the front provides the dumper to keep traction on uneven ground (oscillation).

Operating in pedestrian areas

Before working in a pedestrianised area make sure there is an alternative route for the pedestrians with signs and barriers. Taking into account the machines movements, fumes and noise.

Shut down procedures

On completion of the working day make sure the roller is refuelled to prevent condensation building up in the tank.

- Park roller in a safe place
- Handbrake on and in neutral
- Let engine cool down before switching off engine
- Remove key, lock door and isolate the machine
- Climb off machine (facing dumper 3 points of contact)
- Walk round the machine checking for any damage

Do not park the roller on pedestrian walkways, near trenches, emergency exits or on slopes

Highways

If the machine is being travelled or working on a public highway the road traffic act applies:

- UK driving licence, class B and class
- Aged 21 years for machines exceeding 7.5 tonne

Loading and Unloading from a Transporter

Before loading or unloading check:

- The transporter is capable of taking the weight of the machine
- Firm, level ground for the transporter
- Safe area round the transporter (no personnel walking round)
- No overhead power lines
- Roller is clean
- Ramps of the transporter are in line with the roller
- Banksman to guide operator on

The loading operations is the responsibility of the transporter driver but if the operator is to load the roller on behalf of the transporter driver, they must make sure the roller is in the agreed position on the bed of the transporter, with the handbrake on, in neutral, keys out, door lock